Manufacturing and Processing Technology 1

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Manufacturing and Processing Technology 1 (A11413)

Short Title: Manufac & Process Technology 1
Faculty Science and Computing
Department Science
Cluster Pharmaceutical Science
Author EOWENS
Credits 10 **Level:** Introductory

Description of Module / Aims

This module provides an introduction to the processes, equipment and unit operations within the pharmaceutical industry. It also discusses waste treatment (including water and solvent recovery), as well as the role of packaging within the pharmaceutical industry.

Programmes

MANU-0001	Higher Certificate in Science in Good Manufacturing Practice and Technology (WD_SGMPT_C)	2 / 3 / M

Indicative Content

- Pharmaceutical plants: dedicated vs MPP, batch vs continuous processing
- Synthetic routes, individual chemical process operations, including scale up from laboratory to production
- Route from R&D to approval for new drugs
- Reactors: basic design, control and operation, material of construction, agitation, heating and cooling reactor units
- Recovery and purification of bulk pharmaceutical products: distillation, crystallisation, filtration, centrifugation, drying
- Types & uses of different water grades and purification processes
- Role of packaging in the pharmaceutical industry

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Distinguish between batch and continuous processing, including advantages and disadvantages of each.
2. Discuss the manufacturing trends occurring across the pharmaceutical industry, including MMP vs dedicated plants.
3. Identify and describe the basic unit operations and equipment used in the synthesis and purification of bulk pharmaceuticals.
4. Describe the route from R&D to approval for a new API.
5. Describe basic issues associated with scale-up, process safety, environmental and cost considerations.
6. List the grades of water used in a pharmaceutical plant, identify where each grade of water can be used, and describe purification processes for each grade.
7. Describe chemical and physical treatment of waste and waste water.
8. Discuss the role of packaging in the pharmaceutical industry. Identify the necessary criteria when designing packaging for pharmaceutical products and list the information required on packaging for OTC drugs.

Learning and Teaching Methods

- Lectures.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	5,6,7,8
Continuous Assessment	50%	1,2,3,4
In-Class Assessment	50%	1,2,3,4

Assessment Criteria

<40%: Inability to demonstrate an understanding of the subject matter.
40%-49%: Demonstrate a limited knowledge of the subject matter.
50%-59%: Demonstrate satisfactory knowledge of the main issues of the subject matter.
60%-69%: Demonstrate good knowledge of the subject matter.
70%-100%: Demonstrate knowledge and understanding of the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	12	12
Practical	12	12
Independent Learning	99	99

Essential Material

- Lee, S. and G. Robinson. _Process development:fine chemicals from grams to kilograms_. 1st ed.. Oxford: Oxford University Press, 1995.
- Nusim, S. _Active Pharmaceutical Ingredients: Development, Manufacturing, and Regulation (Drugs and the Pharmaceutical Sciences)_. 2nd Edition. USA: Informa Healthcare, 2009.